

Project Report

Pairwise Epistatic Interactions and Their Functional Consequences in Sorghum

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Abstract

This project explores the potential role of genetic variant interactions (epistasis) in shaping the genetic architecture of complex traits in *Sorghum bicolor*. By analyzing single-nucleotide polymorphism (SNP) data alongside phenotypic information, we aimed to identify significant interactions that influence traits such as Amylose, Fat, Protein, and Grain Yield. By utilizing the MM4LMM R package, we developed a pipeline to analyze millions of SNP pairs across multiple phenotypes to identify significant interaction effects, trying to shed light on how combinations of genetic variants influence the expression of different traits in *Sorghum bicolor*. The project implemented stringent quality control steps, including filtering SNPs based on minor allele frequency(MAF) and linkage disequilibrium(LD), to focus on variants most likely to impact trait variation. Additionally, principal components and a kinship matrix were incorporated to account for population structure and genetic relatedness. Both interaction and marginal effects were explored, with models fitted using Restricted Maximum Likelihood (REML). Although no interactions or marginal effects survived correction for multiple testing, further analyses based on raw p-values provided additional insights. Using exploratory analyses, including 2x2 contingency tables and Chi-squared tests, we evaluated the relationships between marginal and interaction significance at varying thresholds. These tests revealed evidence of enrichment for SNP pairs in some traits(e.g **Plant Height**, **DTA**), while depletion was observed in others(e.g **Fat**, **Grain Yield**). This project underscores the challenges in detecting epistasis in complex traits, particularly under stringent statistical correction thresholds, but also demonstrates the value of exploratory analyses for revealing enrichment and depletion patterns across traits. The developed pipeline offers a scalable framework for epistasis detection and lays the groundwork for integrating larger datasets and functional annotations to better understand the genetic interactions shaping complex traits.

1. Introduction

Genome-wide association studies (GWAS) have significantly enhanced our understanding of the genetic basis of complex traits. These studies have identified thousands of single-nucleotide polymorphisms (SNPs) associated with various phenotypes, providing critical insights into genetic contributions to disease, productivity, and adaptability. However, GWAS traditionally focuses on single-locus analyses, which examine the effects of individual SNPs in isolation. This approach could oversimplify the complex biological mechanisms underlying traits. Genetic interactions between SNPs, known as epistasis, are increasingly recognized as key contributors to trait variability. Epistatic interactions can reveal how combinations of genetic variants influence phenotypes, but their detection poses significant computational and statistical challenges due to the large number of possible SNP combinations in genome-wide datasets.

Sorghum bicolor, commonly known as sorghum, is a vital cereal crop cultivated for its grain and biomass. Its adaptability to diverse and often challenging environmental conditions makes it a critical resource for food security and renewable energy. Sorghum grain serves as a staple food source for humans and animals[1], while its biomass is increasingly utilized in bioenergy production, contributing to sustainable energy initiatives[2]. The sequencing of the Sorghum bicolor genome in 2009 marked a significant milestone, enabling molecular genetic studies and advancing our understanding of its adaptation mechanisms[3]. The Sorghum Association Panel (SAP), a key genomic resource with whole-genome sequencing (WGS) data available, plays a crucial role in capturing the genetic diversity within sorghum breeding programs, providing a valuable foundation for advancing genetic studies and crop improvement efforts[4].

In this project, we set out to investigate the genetic architecture of complex traits in sorghum by analyzing both the individual effects of single SNPs (marginal effects) and the combined effects of SNP pairs (epistatic interactions). In total, 10 traits were investigated in this project. Using phenotypic data from the Sorghum Association Panel (SAP) and corresponding genotypic data, we utilized the MM4LMM R package to examine millions of SNP combinations. Restricted Maximum Likelihood (REML) was employed to fit both interaction and marginal models, incorporating principal components and kinship matrices to account for population structure and genetic relatedness.

To ensure the reliability of the results, quality control steps were applied, including filtering SNPs based on minor allele frequency (MAF) and linkage disequilibrium (LD) and retaining only those located within protein-coding regions. This approach allowed us to focus on high-confidence genetic variants most likely to influence the traits of interest. Both marginal and interaction effects were thoroughly assessed. Although no significant associations were identified after adjusting for multiple testing, additional exploratory analyses provided valuable context. By leveraging Chi-squared tests and 2x2 contingency tables, we examined patterns of enrichment across different significance thresholds, uncovering probable preliminary relationships between marginal and interaction effects.

We hypothesize that epistatic interactions play an important role in shaping the genetic architecture of complex traits in sorghum. However, the detection of such interactions is inherently challenging due to the stringent thresholds required for statistical significance and the computational complexity of evaluating millions or even billions of SNP combinations. We predict that while significant epistatic interactions might not readily emerge under these constraints, exploratory analyses will reveal patterns of relationships

between marginal and interaction effects, offering preliminary insights into the role of genetic interactions in trait variability.

To test this hypothesis, we applied a systematic workflow to investigate both marginal SNP effects and SNP-SNP interactions across traits, incorporating statistical methods to account for population structure and genetic relatedness. Our project explores the extent to which significant relationships can be detected and evaluates trends in the distribution of raw p-values to identify potential biological signals.

2. Materials and Methods

2.1 Data available

The data used in this project includes both genotypic and phenotypic datasets derived from the **Sorghum Association Panel (SAP)**, a diverse collection of 400 sorghum accessions widely used in genetic studies.

Genotypic Data:

Single-nucleotide polymorphism (SNP) data were obtained in **VCF (Variant Call Format)** directly from **SAP.imputed.vcf(15.189.381 total variants)**. These SNPs represent genetic variants distributed across the Sorghum bicolor genome. The dataset provides a comprehensive view of the genetic diversity within the SAP and serves as the basis for analyzing both marginal effects and SNP-SNP interactions.

Phenotypic Data:

The phenotypic dataset contained measurements for the following traits, each selected for its agricultural importance:

1. **Amylose:** A key component of starch that influences grain texture, digestibility, and quality.
2. **Fat:** A trait affecting the nutritional composition and energy content of the grain.
3. **Protein:** Critical for grain quality and nutritional value, influencing food and feed applications.
4. **Grain Yield (GY):** The total production of grain, a primary productivity indicator for crop improvement.
5. **Cal.g:** The caloric content per gram, representing the energy density of the grain.
6. **DTA (Days to Anthesis):** The number of days to flowering, a measure of crop maturity and adaptation to different environments.
7. **PH (Plant Height):** A structural trait associated with biomass yield, lodging resistance, and overall plant architecture.
8. **GN (Grain Number):** The total number of grains per plant, an important yield-related trait.
9. **GW (Grain Weight):** The average weight of individual grains, contributing to overall grain yield.
10. **Starch:** A major carbohydrate component of the grain.

The phenotypic data were provided in **CSV format(SAP.pheno.csv)**. These traits reflect critical aspects of sorghum productivity, quality, and adaptation. Out of the original 400

accessions present in the dataset, 13 were filtered out since phenotypic data was not available for the traits listed above.

2.2 Data Preprocessing

To ensure high-quality input data and robust analyses, several preprocessing steps were applied to the genotypic data.

Initial Genotypic Data Preparation

- **SNP Naming:**
The starting VCF file from the Sorghum Association Panel (SAP) dataset was updated by assigning unique identifiers to SNPs in the format **CHR:POS** (chromosome and position). This standardization resulted in downstream identification and mapping of SNPs.
- **Biallelic SNP Filtering:**
To ensure data quality, **bcftools** was used to filter the VCF file, retaining only **biallelic SNPs** (variants with exactly two alleles: reference and alternate). This step eliminates multiallelic variants, which are more complex to model and less commonly used in epistasis analyses.
- **CDS Region Filtering:**
SNPs were further filtered to retain only those located within **coding sequences (CDS)**. This was achieved using the **Sorghum_bicolor.Sorghum_bicolor_NCBIv3.55.gff3** annotation file, which provided genomic coordinates for the CDS regions. Focusing on the CDS ensures that the analysis targets variants most likely to influence functional changes in proteins, which are more relevant for studying phenotypic traits.
- **Accession Matching:**
Thirteen accessions that did not have corresponding phenotypic data were removed to ensure consistency between the genotypic and phenotypic datasets. This step guaranteed that only individuals with complete phenotypic information were retained for downstream analyses.

The filtered genotypic dataset was then used to generate two separate VCF files for distinct purposes:

File 1: VCF for Kinship Matrix Calculation

- **Minor Allele Frequency (MAF) Filtering:**
SNPs with a **MAF ≥ 0.05** were retained to exclude rare variants that could introduce noise or instability into the kinship matrix calculation. This threshold ensures that retained SNPs are common enough to contribute meaningful genetic signals.
- **Kinship Matrix Generation:**
The MAF-filtered VCF file was processed using **TASSEL 5**, a widely used tool for genetic analysis, to calculate the **kinship matrix**. The kinship matrix models genetic relatedness between accessions, which is crucial for

accounting for non-independence in mixed models and ensuring accurate estimates in the MM4LMM framework.

File 2: VCF for Epistatic and Marginal Effects Analysis

- **MAF Filtering:**
SNPs with a **MAF ≥ 0.25** were retained to focus on variants with moderate-to-high allele frequencies, which are more likely to exhibit detectable epistatic effects in genome-wide analyses.
- **Linkage Disequilibrium (LD) Pruning:**
The MAF-filtered VCF underwent **LD pruning**(using PLINK 2.0[8]) to reduce redundancy among highly correlated SNPs. LD pruning was performed using the following parameters:
 - **Window size:** 100 kb
 - **Step size:** 50 kb
 - **r^2 threshold:** 0.2

The **r^2 threshold of 0.2** was used to quantify the extent of linkage disequilibrium between pairs of SNPs within each 100 kb window. The r^2 value represents the correlation between two SNPs, with values closer to 1 indicating higher correlation and values closer to 0 indicating independence. During the pruning process, PLINK 2.0 calculated the r^2 value for all SNP pairs within the specified window. If the r^2 value between two SNPs exceeded the threshold of 0.2, one SNP was removed, ensuring that the retained SNPs were largely independent of each other. The window was then shifted by 50 kb (step size) to include overlapping regions, and the process was repeated. By retaining only independent SNPs, we optimized the dataset for the detection of epistatic interactions, reducing redundancy without compromising the genomic representation required for reliable analyses of the SAP accessions.

These parameters were selected based on the observed **LD decay in sorghum**. As shown in **Figure 1**[4], LD (r^2) decreases rapidly over short distances and stabilizes below 0.2 within approximately **100–150 kb** across the genome and individual chromosomes. By using a window size of 100 kb and an r^2 threshold of 0.2, we ensured that only largely independent SNPs were retained for downstream analyses. This approach minimizes collinearity, reduces computational burden, and improves the reliability of epistatic interaction detection.

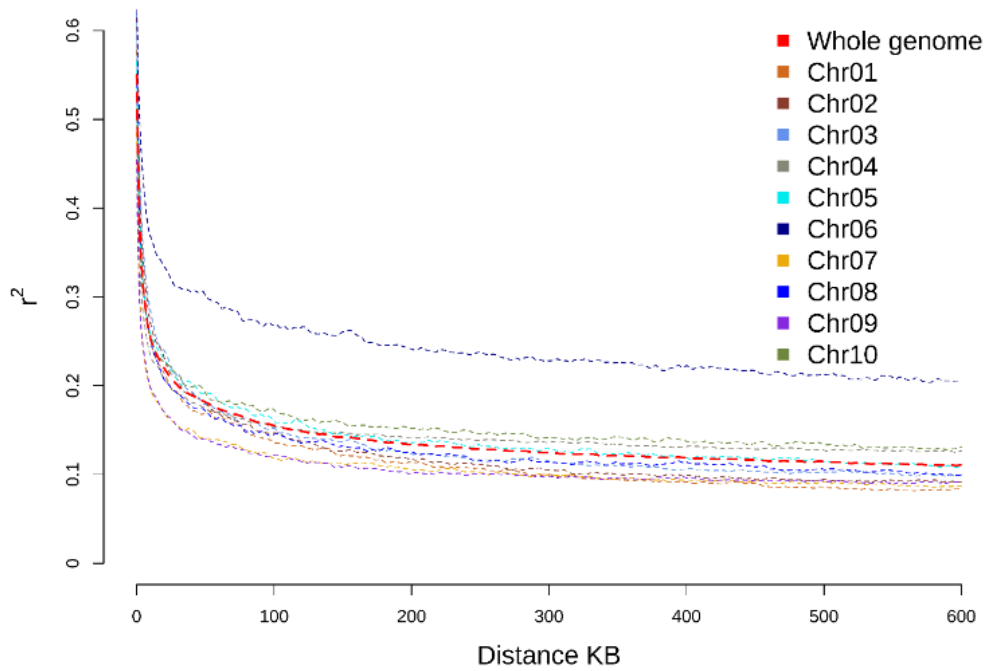


Figure 1. Linkage disequilibrium decay across each sorghum chromosome and across the genome(Boatwright et al.[4] Supplementary Material)

The original SAP dataset contains **15.189.381** variants and after preprocessing we are left with **2735** variants. This means that we have **2735 choose 2 = 3.738.745** possible SNP combinations for downstream epistatic analysis and **2735** SNPs for marginal analysis for **387** sorghum accessions.

Principal Component Analysis (PCA):

Principal Component Analysis (PCA) was performed on the SNP dataset filtered with a Minor Allele Frequency (MAF) threshold of 0.05 using PLINK 2.0[8]. This file was chosen as it provides a comprehensive set of common variants while avoiding noise from rare variants, which could disproportionately influence the eigenvalues. PCA identifies the main axes of genetic variation, capturing population structure within the dataset.

The first three principal components (PCs) were retained, as they exhibited the highest eigenvalues and cumulatively explained almost half of the total genetic variance. These components are likely to capture the majority of stratification within the SAP, effectively modeling population structure without overfitting the mixed models.

The percentage of variance explained for the first three PCs were as follows: **PC1 = 20.60%**, **PC2 = 16.79%**, and **PC3 = 11.35%**, cumulatively explaining **~49%** of the total variance. Subsequent PCs explained substantially less variance and were not included, as they were unlikely to provide meaningful contributions to the models.

2.3 FDR Correction for Multiple Testing

Given the large number of hypotheses tested for both marginal SNP effects and SNP-SNP interactions, the raw p-values were adjusted for multiple testing using the False Discovery Rate(FDR) correction to control for type I errors[5]. FDR correction is a widely used approach, especially in studies involving large-scale hypothesis testing as it balances the discovery of true positives with the risk of false positives.

The Benjamini-Hochberg (BH) [6] procedure was employed to compute FDR-adjusted p-values. This procedure begins by ranking all individual p-values in ascending order, with the smallest p-value assigned a rank of 1, the second smallest a rank of 2, and so on. Each p-value is then compared against a critical threshold calculated using its rank, the total number of tests, and a predefined False Discovery Rate, typically set as a percentage like 5%. The procedure identifies the largest p-value that is smaller than its corresponding critical value, and all p-values smaller than this are considered significant. This systematic ranking and comparison help balance the discovery of true associations with the control of false positives, making it an effective method for handling multiple testing.

2.4 Enrichment Analysis

Building on the marginal and epistatic analyses, the enrichment analysis was conducted to explore whether SNP pairs with marginally significant effects were overrepresented among those with significant interaction effects. This analysis provided insights into the potential dependencies between marginal and interaction significance.

The enrichment analysis was structured around 2x2 contingency tables constructed for each phenotype and significance threshold. For each phenotype and threshold, 2x2 contingency tables were constructed to categorize SNP interactions based on two criteria:

1. Marginal Significance: Whether at least one SNP in the interaction pair was significant
2. Interaction Significance: Whether the interaction term itself was significant

The resulting table classified interactions into four categories:

- TRUE/TRUE: Interactions with at least one marginally significant SNP and a significant interaction effect
- TRUE/FALSE: Interactions with at least one marginally significant SNP but no significant interaction effect.
- FALSE/TRUE: Interactions with no marginally significant SNPs but a significant interaction effect.
- FALSE/FALSE: Interactions with neither marginally significant SNPs nor a significant interaction effect.

For all the phenotypes and thresholds, observed counts were calculated, particularly for the TRUE/TRUE cell, representing SNP pairs with both marginal and interaction significance.

To evaluate whether the observed count in the TRUE/TRUE cell showcased enrichment, the expected count was calculated under the assumption of independence between marginal and interaction significance. This independence model assumes that the marginal significance of SNPs is unrelated to their involvement in significant interactions. The expected count of the TRUE/TRUE cell was derived using the following formula:

$$Expected\ Count(TRUE/TRUE) = \frac{Row\ Total(Marginal\ TRUE) \times Column\ Total(Interaction\ True)}{Grand\ Total}$$

The enrichment ratio was then computed as:

$$Enrichment\ Ratio = \frac{Observed\ Count(TRUE/TRUE)}{Expected\ Count(TRUE/TRUE)}$$

An enrichment ratio greater than one indicates that significant SNP-SNP interactions are overrepresented among SNP pairs with marginally significant SNPs.

3. Results

3.1 R Pipeline for Epistasis Analysis

To facilitate a reproducible and modular framework for the analysis of SNP-SNP interactions in the SAP, we designed an R pipeline using the **targets** package. The pipeline integrates data preprocessing, statistical model fitting(by **MM4LMM**), and results extraction, enabling efficient analysis of a large dataset. Below we describe the key components of the pipeline.

3.1.1 Data Loading and Preprocessing

The pipeline begins by loading and preprocessing genotypic, phenotypic and auxiliary data.

1. Loading and Processing Genotypic Data

The genotypic data, provided in VCF format, is loaded using the **vcfR** package. From the VCF file, genotype information is extracted for all SNPs retained after filtering for MAF and LD pruning(File 2 in Data Preprocessing). Genotype values are converted from categorical representations(e.g 0|0,1|0) into numerical values(0,1,2), representing the number of alternate alleles. This transformation enables the downstream statistical model to interpret genotypes as continuous variables.

2. Principal Components(PCs) and Kinship Matrix

The principal components and kinship matrix are loaded into the pipeline since they were computed outside of it.

3. Phenotypic Data

The phenotypic data is loaded and filtered to ensure consistency with the genotypic dataset. Only accessions present in the genotype matrix are retained, ensuring alignment between the two datasets. The phenotypic values are then matched to the genotypic data for subsequent analysis.

3.1.2 SNP Pair Generation and Interaction Terms

To evaluate SNP-SNP interactions, the pipeline generates all possible pairwise combinations of SNPs from the pruned dataset. For each SNP pair, an interaction term is computed as the product of the additive genotype codes (0, 1, 2) at both SNPs in the pair. For example if **SNP1** is 1(0|1 or 1|0) and **SNP2** is 2(1|1) then their product would be **SNP1 X SNP2** = 1 X 2 = 2(interaction term). These interaction terms represent additive x additive (AxA) interactions, which quantify how the combined effects of two SNPs on a trait deviate from the sum of their independent (additive) effects.

It is important to note that AxA interactions capture non-additive effects. While the interaction terms are derived using additive genotype codes, they model how the combined influence of two SNPs differs from the expectation based solely on their additive contributions. This distinction undermines that SNP-SNP interactions reflect non-additive genetic relationships, where the effect of one SNP depends on the presence or absence of another.

Given the large number of SNPs, this step leverages parallel processing via the **future.apply** package to optimize computational efficiency across available cores.

3.1.3 Statistical Model Fitting

The pipeline uses the **MM4LMM** package to fit linear mixed models for epistasis analysis. The models are constructed as follows:

1. **Response Variable:**
The phenotypic trait serves as the dependent variable.
2. **Fixed Effects:**
Fixed effects include the genotypes of the two SNPs (additive effects), the interaction term (epistatic effect), and the first three PCs (to account for population structure).
3. **Random Effects:**
The additive genomic value is incorporated as a random effect through the kinship matrix to account for genetic relatedness among accessions.

The model fitting process is parallelized to efficiently handle the large number of SNP pairs (NbCores, built in the package). The pipeline utilizes a restricted maximum likelihood (REML) approach for parameter estimation.

3.1.4 Results Extraction and Summarization

After fitting the models, the pipeline extracts key statistics for each SNP pair, including:

- **Beta Coefficients:** Estimates for the additive effects of SNP1 and SNP2, as well as their interaction term.
- **Standard Errors (SE):** Derived from the variance-covariance matrix for SNP1, SNP2, and the interaction term.
- **SNP Identifiers:** The genomic coordinates of the SNPs involved in each interaction.

These results are compiled into a structured matrix, enabling easy interpretation and further analysis. The pipeline also handles defective entries (e.g., improperly formatted data or

invalid VarBeta matrix dimensions), skipping them and reporting their counts to ensure data quality.

3.1.5 Integration and Automation

The pipeline integrates all steps into a modular workflow managed by the **targets** package. This framework ensures that:

- Dependencies between steps are explicitly defined.
- Intermediate outputs are cached for efficiency.
- The pipeline can be easily reproduced or updated as needed.

Each step is encapsulated in a separate function, allowing for easy debugging and modification. The workflow is initialized via the **_targets.R** script, which defines the targets and specifies the order of execution. Key targets include:

- **vcf_data**: Genotypic data extracted from the VCF file.
- **geno_numeric**: Numeric genotype matrix.
- **pcs_3**: First three PCs extracted from the genotype data.
- **phenotype_data**: Filtered phenotypic dataset.
- **filtered_kinship_matrix**: Kinship matrix aligned with the genotypic data.
- **X_triplet_list**: SNP pairs and interaction terms.
- **reml_full_3pcs**: Results from fitting the REML models.
- **result_df**: Final extracted results for all SNP pairs.

The pipeline leverages parallelization via the **future.apply** package and optimizes resource usage by distributing tasks across multiple cores. Additionally, intermediate outputs are saved in **.qs** format(qs library) for faster read/write operations compared to the standard **.rds** format.

3.2 Association testing

3.2.1 Raw p-value distribution for Interaction effects

As a first step in our analysis, we explored the raw p-value distributions to understand their shape and what they reveal about SNP interactions across different phenotypes. P-values often exhibit distinct patterns, such as an L-shaped distribution when there is enrichment of significant interactions, or a uniform distribution when there is no clear association. By examining these distributions, we aimed to evaluate if there were any patterns suggesting meaningful interactions or if the results are more consistent with random noise.

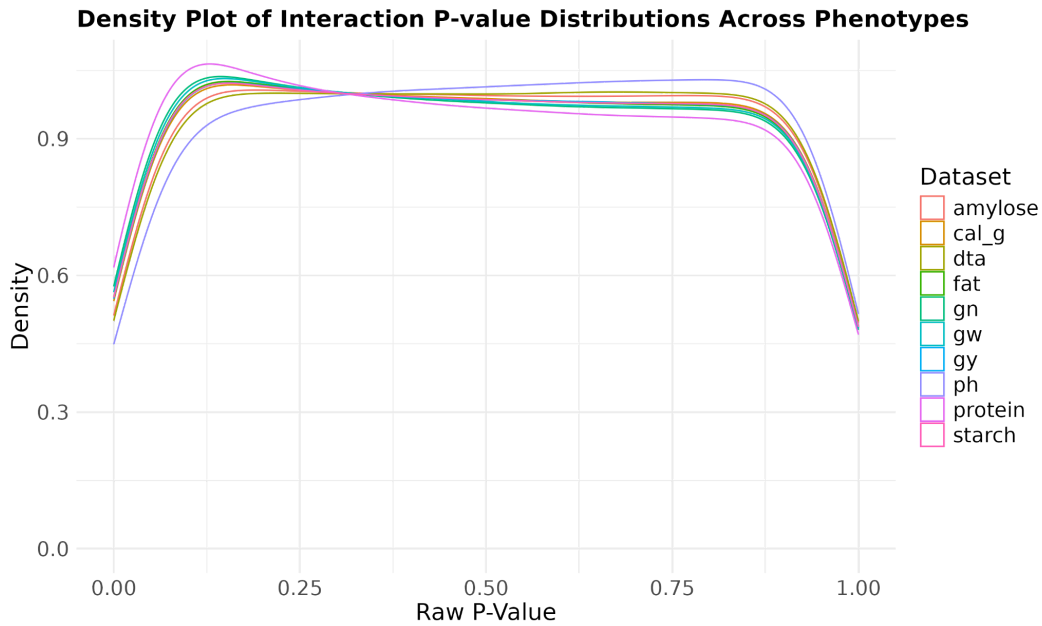


Figure 2. Density Plot of P-Value Distributions Across Phenotypes. This figure shows the distribution of raw p-values for SNP-SNP interactions across ten phenotypes. Each line represents the density curve for a specific phenotype, capturing the shape of its p-value distribution.

To examine these raw p-value distributions, we visualized them using density plots. The density plot in Figure 1 provides a detailed view of how the p-values are distributed, allowing us to compare phenotypes and identify potential trends. The x-axis represents the range of p-values (from 0 to 1) and the y-axis represents the estimated density, which reflects the relative concentration of p-values in the different parts of the range. It's important to note that the y-axis shows how densely p-values are distributed, with the area under each curve summing to 1.

As we can see in this density plot, phenotypes such as **Protein** and **Grain Number(gn)**, display noticeable peaks near 0, suggesting a higher density of p-values in the range [0,0.1]. Such peaks might indicate an enrichment of SNP-SNP interactions for these phenotypes. In contrast, phenotypes like **Amylose** and **DTA** show a flatter, more uniform distribution, which is often associated with random patterns and lack of strong associations.

3.2.2 Raw p-value Distribution for Marginal Effects

While interaction effects are key to understanding the combined influence of SNP pairs on phenotypes, it is also important to explore marginal effects, which focus on the independent contribution of each SNP. Examining both types of effects together provides a more complete understanding of how genetic variation contributes to each trait and helps reveal relationships between individual SNPs and their combined effects.

When one or both SNPs have significant marginal effects, it suggests that these SNPs independently influence the phenotype. If their interaction effects is also significant, it reinforces the idea that these SNPs work both individually and together to affect the trait, possibly in a (anti-)synergistic way. However, if the marginal effects are significant but the interaction is not, it could indicate that the SNPs primarily act independently, and their combined effect does not contribute additional information about the phenotype.

On the other hand, a significant interaction effect in the absence of significant marginal effects indicates a more complex relationship. This pattern suggests that the SNPs do not have strong individual effects but work together in a way that impacts the phenotype. Such interactions could reflect epistasis, where the effect of one SNP depends on the presence or absence of the other. This type of result highlights the importance of analyzing interactions, as significant relationships between SNPs can exist even when neither SNP is independently associated with the phenotype.

In order to examine the role of marginal effects, we visualized the raw p-value distributions for each phenotype in a density plot, like we did with the raw interaction p-values in Figure 1.

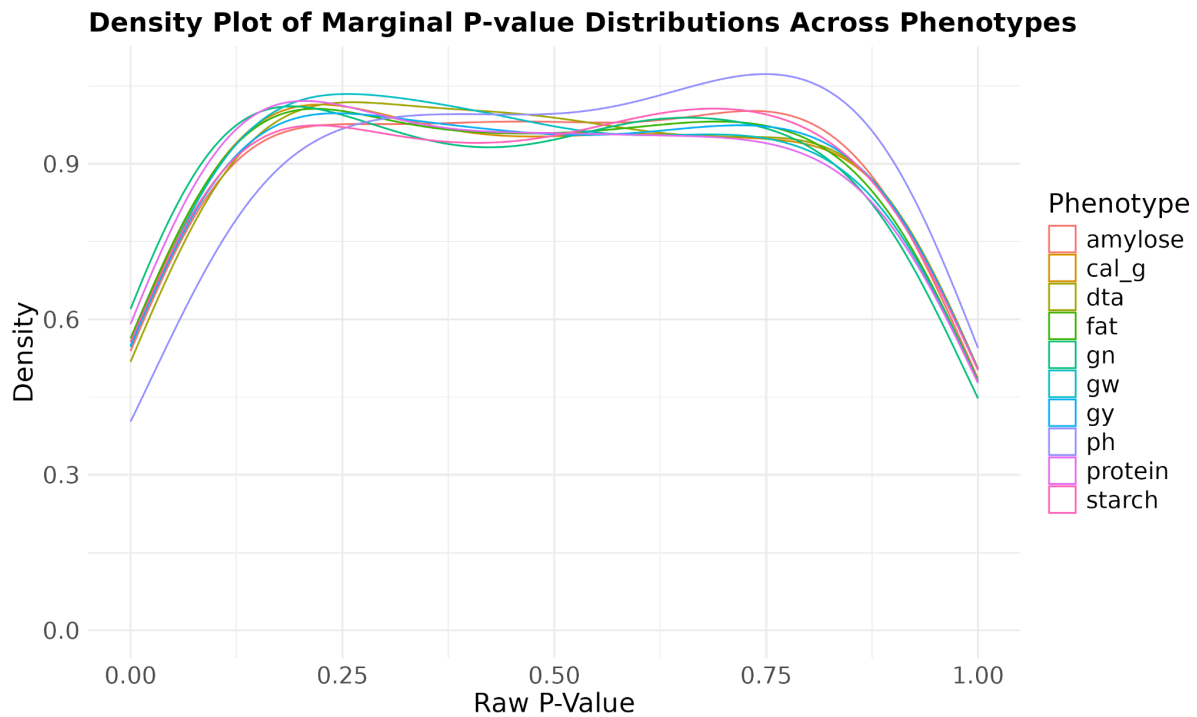


Figure 3. Density Plot of Marginal P-value Distributions Across Phenotypes. This plot shows the density of raw marginal p-values for all phenotypes analyzed. Each colored line corresponds to a specific phenotype, as shown in the legend on the right-hand side of the plot. Higher peaks reflect a higher frequency of SNPs with p-values in that range, while flatter regions suggest a more uniform distribution of p-values.

From the plot, the majority of the phenotypes exhibit deviations from the typical uniform distribution of p-values expected under the null hypothesis. Traits like **Grain Number (gn)** and **Protein** show an enrichment of small p-values, indicating the presence of true genetic signals and suggesting these phenotypes are influenced by significant genomic factors. In contrast, phenotypes like **PH** and **DTA** exhibit peaks toward the higher p-value range, suggesting weaker genetic associations or lower detectability of significant effects. These patterns may reflect differences in the genetic architecture, heritability, or statistical power for each phenotype.

It is also worth noting that the strict filtering applied at the beginning of the whole pipeline including biallelic SNP filtering, CDS filtering, MAF and LD pruning, may have shaped these distributions. While such filtering ensures high-quality data and computational feasibility, it can also reduce the ability to detect weaker genetic signals. This is particularly

relevant given that no significant effects passed the FDR correction, despite evidence from other studies, such as Boatwright et al.[4], which demonstrated the persistence of GWAS peaks for certain traits even under stringent filtering.

3.2.3 FDR Correction for Multiple Testing

After applying the Benjamini-Hochberg procedure to control for $FDR(<0.05)$, none of the tested SNP-SNP interactions or marginal effects passed the FDR threshold for statistical significance across any phenotype. The lack of significant results after FDR correction likely reflects the limited power of the project to detect small genetic effects in the context of stringent multiple testing corrections. This limitation arises from the small effect sizes typical of polygenic traits, the large number of hypotheses tested, and the strict FDR threshold applied. Future studies with larger datasets, integration of functional annotations, or alternative statistical approaches could help overcome these challenges and uncover meaningful genetic interactions.

3.2.4 Posterior Inclusion Probabilities(PIPs)

The absence of significant results after applying FDR correction highlighted the challenges of detecting small genetic effects in the context of polygenic traits, where stringent statistical thresholds and high dimensionality often obscure true associations. To address this limitation, we explored Posterior Inclusion Probabilities (PIPs) using an empirical Bayes framework as an alternative statistical approach.

PIPs quantified the probability that a given SNP-SNP interaction contributes to the phenotypic variation, integrating both the observed data and prior expectations about effect sizes. This approach provides a more nuanced measure of evidence for association compared to p-values, which are limited to hypothesis testing under the null hypothesis. By ranking SNP-SNP interactions based on their PIPs, we were able to highlight potential candidates that might have been overlooked due to low statistical power or stringent significance thresholds.

To get the significant results we chose a threshold of **PIP ≥ 0.8** . Intuitively, this means there is an 80% probability that the interaction is part of the underlying genetic model influencing the trait, based on both the observed data and prior assumptions. Unlike p-values, which test the plausibility of the null hypothesis, PIPs directly quantify the likelihood that an interaction is truly associated with the phenotype. While PIPs don't definitively prove the interaction is true, it highlights SNP pairs as strong candidates for further validation or investigation.

As we can see from Figure 4 most traits exhibit a small number of SNP interactions that surpass the 0.8 threshold. The exception to this is **Grain Weight(gw)** and **Protein** which both show a very high number of possibly significant interactions. Specifically for each trait we have:

Table 1. Number of PIPs >0.8 for each trait.

Trait	No. of PIPs > 0.8
Amylose	0
Cal.g	6

DTA(days to anthesis)	0
Fat	1
Grain Numer(gn)	29
Grain Weight(gw)	961
Grain Yield	6
Protein	996
Starch	0
Plant Height(PH)	N/A

The large number of high-PIP candidates for **Protein** and **Grain Weight** likely reflects a combination of their genetic complexity, stronger signals in the data, and compatibility with the statistical model used. These traits may provide more opportunities to uncover meaningful genetic interactions compared to traits with weaker signals. However, further validation is needed as to why they exhibit such a high number of significant PIPs.

On the other hand **Amylose** and **DTA** showcase a rather strange behaviour when comparing PIPs to p-values. For example all **Amylose** interactions show that all PIPs are within [0.77,0.78] no matter how high or low the corresponding p-value is. **DTA** follows the same pattern. This clustering of PIPs suggests that the Bayesian framework did not identify any strong signals for SNP-SNP interactions in these traits. The uniformity across p-values indicates that the p-values alone are insufficient to provide meaningful updates to the prior probabilities, leading to nearly identical posterior probabilities(Supplementary Figure 1).

The only phenotype that didn't generate any PIPs was **PH**(plant height). It is also the only case that exhibits a p-value distribution that starts low on the left end and rises higher on the right end. This led us to believe that the Bayesian method utilized, interprets this pattern as a lack of evidence to update the prior probabilities, resulting in PIPs of 0 for all SNP pairs.

While posterior inclusion probabilities offer a powerful method for identifying SNP-SNP interactions, some potential limitations should be considered. One issue is the high dimensionality of genomic data, which can lead to unreliable PIP estimates, particularly for loci with small effect sizes or low posterior probabilities[9][10]. Since the **ashr** package was used to calculate PIPs, the prior distributions for effect sizes were estimated empirically from the observed data using a flexible mixture of normal distributions. While LD pruning was applied to reduce redundancy among SNPs, weak residual LD may still influence PIPs in certain genomic regions. When SNPs are in LD, the PIPs may reflect the combined effects of multiple loci rather than the individual contributions of each locus to the phenotypic variation[11]. This can make it challenging to disentangle the specific roles of individual SNPs in epistatic interactions, leading to potential misinterpretations of the genetic architecture of traits[11].

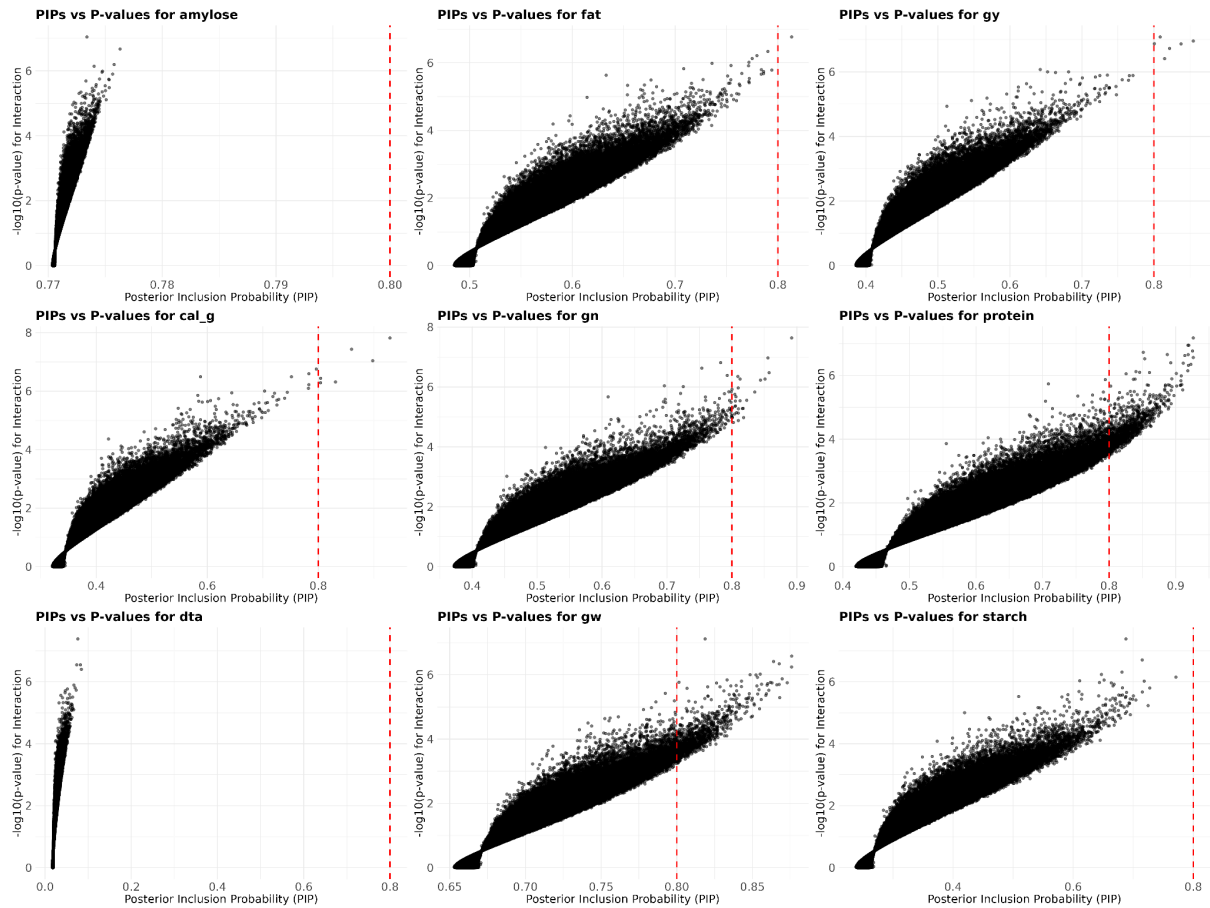


Figure 4. PIPs against log-transformed p -values for all traits (Amylose, Fat, Grain Yield, Cal.g, Grain Number, Protein, DTA, Grain Weight, Starch). The red line indicates the 0.8 threshold.

3.3 Enrichment Analysis

The enrichment analysis aimed to assess the relationship between marginal SNP effects and SNP-SNP interaction effects across the phenotypes, by leveraging contingency tables and statistical tests. The analysis explored whether significant interaction effects are enriched among SNP pairs involving marginally significant SNPs.

3.3.1 Statistical Testing

To validate the reliability of the expected counts and assess whether the observed counts deviated significantly from what would be expected under the independence model, statistical testing was performed on each contingency table.

The primary statistical test used was the chi-squared test of independence, which evaluates whether the observed distribution of counts across the table aligns with the expected distribution under the null hypothesis. The resulting p -value indicates the likelihood of observing the given data (or more extreme data) if the null hypothesis of independence were true. A significant p -value (e.g. $p < 0.05$) suggests that the observed counts deviate significantly from the expected counts, indicating a potential dependency between marginal and interaction significance.

However, the chi-squared test has specific assumptions that must be met for the results to be valid:

1. All expected counts must be greater than or equal to 1.
2. No more than 20% of the cells in the table should have expected counts less than 5.

These assumptions were met for all traits and all thresholds except for **Amylose** at the 0.001 threshold, where expected counts in the contingency table were too low to satisfy the assumptions. In this case, a Fisher's Exact Test was used as an alternative, as it is more appropriate.

The results of the statistical tests, including p-values, test statistics, and degrees of freedom, were stored alongside the enrichment ratios. These results served as a supplementary validation step, providing additional context for interpreting the observed enrichment patterns.

3.3.2 Threshold Sensitivity Analysis

To explore how enrichment patterns varied with the stringency of the significance criteria, the analysis was repeated across four p-value thresholds: 0.05, 0.01, 0.005, and 0.001. By systematically lowering the threshold, the analysis assessed whether stronger enrichment signals emerged at stricter significance levels, which might indicate stronger relationships between marginal and interaction significance.

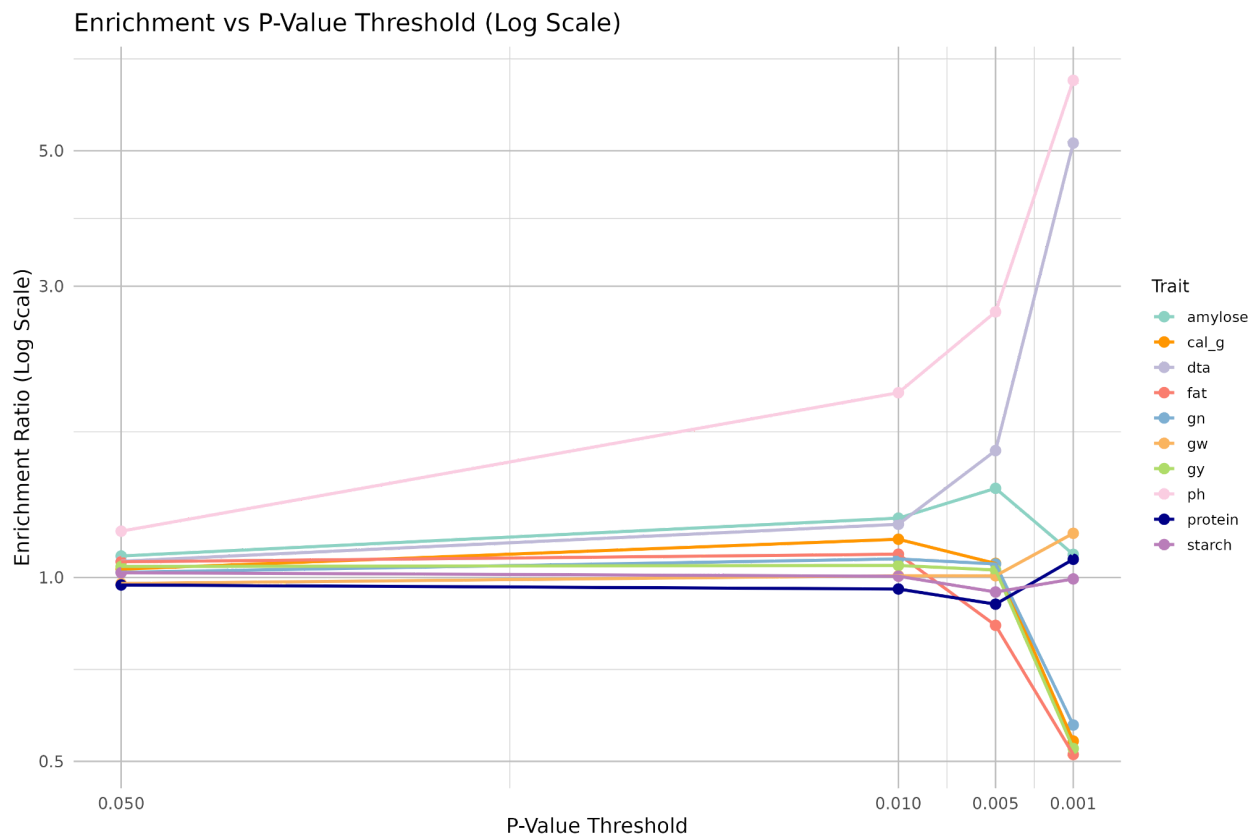


Figure 5. Enrichment Ratios Across P-Value Thresholds for Different Phenotypes
This plot illustrates the enrichment ratios for significant SNP-SNP interactions involving marginally significant SNPs across multiple phenotypes and p-value thresholds. The x-axis represents the p-value thresholds, with stricter thresholds on the right. The y-axis shows the enrichment ratio, which quantifies the overrepresentation of significant SNP-SNP interactions among marginally significant

SNP pairs. Each colored line corresponds to a different phenotype, as indicated in the legend on the right-hand side of the plot.

The results of the enrichment analysis are summarized in Figure 6, which displays enrichment ratios for all phenotypes across the four thresholds. Each line in the plot represents a single phenotype, with enrichment ratios plotted against reversed p-value thresholds (from less stringent to more stringent) for better clarity.

At higher thresholds (e.g. $p=0.05$), enrichment ratios for most phenotypes hovered around 1, suggesting no significant overrepresentation or underrepresentation of SNP-SNP interactions involving marginally significant SNPs. However, at the thresholds become stricter, more distinct patterns emerged for certain traits:

- **Enrichment:** Some phenotypes, such as **Plant Height** and **DTA**, exhibit rising enrichment ratios, with the enrichment becoming more pronounced at the strictest thresholds. This indicates that significant SNP-SNP interactions for these traits are more likely to involve marginally significant SNPs. These trends suggest a stronger dependency between marginal and interaction significance at lower p-values, potentially reflecting biologically meaningful relationships.
- **Depletion:** Interestingly, some phenotypes exhibit depletion at stricter thresholds, with enrichment ratios dropping below 1.0. This depletion implies that significant SNP-SNP interactions for these traits are less likely to involve marginally significant SNPs, suggesting that interaction effects may arise independently of marginal effects. Depletion is equally important to note as it highlights potential mechanisms where interaction effects dominate, independent of individual SNP effects.

The contrasting trends of enrichment and depletion are indicative of differing genetic architectures across traits. Traits with high enrichment at strict thresholds suggest that marginally significant SNPs may contribute directly to interaction effects, potentially reflecting synergistic genetic mechanisms. Similarly, traits showing depletion may also be driven by additive x additive (AxA) interactions, where the effects of SNP combinations are interdependent and cannot be explained by the simple sum of their individual additive contributions. It is important to note that while the tested SNP-SNP interactions are classified as AxA interactions due to their derivation from additive genotype codes, these interactions are inherently non-additive because they model the combined effect of SNPs that goes beyond their independent contributions.

While the enrichment and depletion trends observed in this analysis provide valuable insights, it is important to acknowledge the limitations of this exploratory analysis. The results depend heavily on the significance thresholds used, which can influence the patterns we observe. They're also affected by the size of the dataset and the filtering steps applied during preprocessing. Another important point is that these patterns don't yet have direct biological validation, so they should be viewed cautiously. Future work could include functional annotations or experiments to confirm if the enrichment or depletion reflects real biological processes.

In conclusion, this analysis demonstrates the potential of enrichment ratios to uncover relationships between marginal and interaction significance. The observed patterns, particularly at stricter thresholds, offer initial insights into the genetic architecture of complex traits and provide a basis for possible future investigations.

4. Discussion

The analysis of pairwise epistatic interactions in *Sorghum bicolor* presented in this project highlights both the opportunities and challenges in exploring the genetic architecture of complex traits. Despite the lack of statistically significant findings, this project offers valuable insights into the genetic mechanisms underlying trait variability and lays the groundwork for future research.

A critical aspect of epistasis detection lies in the choice of statistical metrics[7]. In this project, raw p-values offered an initial exploratory look at potential interactions but didn't survive under stringent corrections such as the false discovery rate (FDR). FDR is an essential tool to control false positives in large-scale analyses, yet its strict thresholds can limit sensitivity, particularly for polygenic traits where effect sizes are small. Bayesian posterior inclusion probabilities (PIPs) presented an alternative perspective, emphasizing the likelihood of SNP interactions contributing to phenotypic variation. While PIPs showed promise in identifying candidates missed by p-value thresholds, their reliance on priors requires careful calibration, especially when applied to traits with weaker genetic signals. These findings emphasize the importance of considering multiple metrics and their trade-offs when interpreting results in epistasis studies.

One notable challenge was the statistical power of the project. The dataset, derived from the Sorghum Association Panel, was valuable but limited in sample size, restricting the ability to detect small effect sizes characteristic of epistatic interactions. Larger datasets are crucial to improving power and uncovering more subtle signals. Collaborative efforts to expand the sample size or meta-analyses across similar datasets could provide the scale necessary to overcome these limitations. Additionally, integrating functional annotations, such as regulatory or coding region information, could help prioritize biologically meaningful interactions, enhancing both the interpretability and relevance of detected signals.

It is also really important to note that the strict filtering criteria, including CDS region filtering, minor allele frequency (MAF) thresholds and linkage disequilibrium (LD) pruning, were crucial for ensuring data quality and computational feasibility. However, these steps may also have excluded variants with weaker or rare effects that contribute to epistasis. Exploring alternative filtering approaches or more flexible algorithms could help strike a better balance between computational efficiency and inclusivity in future analyses.

The computational pipeline developed in this project is both scalable and modular, effectively managing the complexity of SNP-SNP interaction testing across multiple traits. However, there are several ways the pipeline could be refined to enhance its efficiency and flexibility. One key improvement would be to incorporate the computation of the kinship matrix and principal component analysis (PCA) directly into the pipeline, rather than computing them externally. Similarly, integrating steps for filtering the initial VCF file—such as minor allele frequency (MAF) thresholds and linkage disequilibrium (LD) pruning—into the pipeline would streamline the workflow, reducing the need for external preprocessing. While the pipeline already leverages CPU-based parallelization to manage the computational load, this parallelization was specifically designed for the GenomeDK cluster. Refining it to be more universally adaptable across different computational environments could improve its usability. Although GPU acceleration is not supported by the packages used, further optimization of the existing CPU-based approach could help manage even larger datasets more efficiently. These enhancements would make the pipeline more self-contained and accessible, while also improving its scalability for diverse research settings.

The enrichment analysis provided valuable insights into the relationship between marginal SNP effects and SNP-SNP interactions, offering guidance for prioritizing SNP-SNP interaction tests in future studies. The observed patterns suggest that traits exhibiting enrichment at stricter thresholds may benefit from focusing on SNP pairs where at least one SNP has marginal significance. Conversely, for traits showing depletion, prioritizing SNP pairs without marginally significant effects may uncover non-additive mechanisms driving phenotypic variability. Validating these trends requires further biological and statistical investigations, such as integrating functional annotations to assess whether the identified SNPs and their interactions align with known regulatory or coding regions. Additionally, expanding datasets or employing more sensitive statistical models could help detect subtle but biologically meaningful interactions, enhancing the importance of these findings.

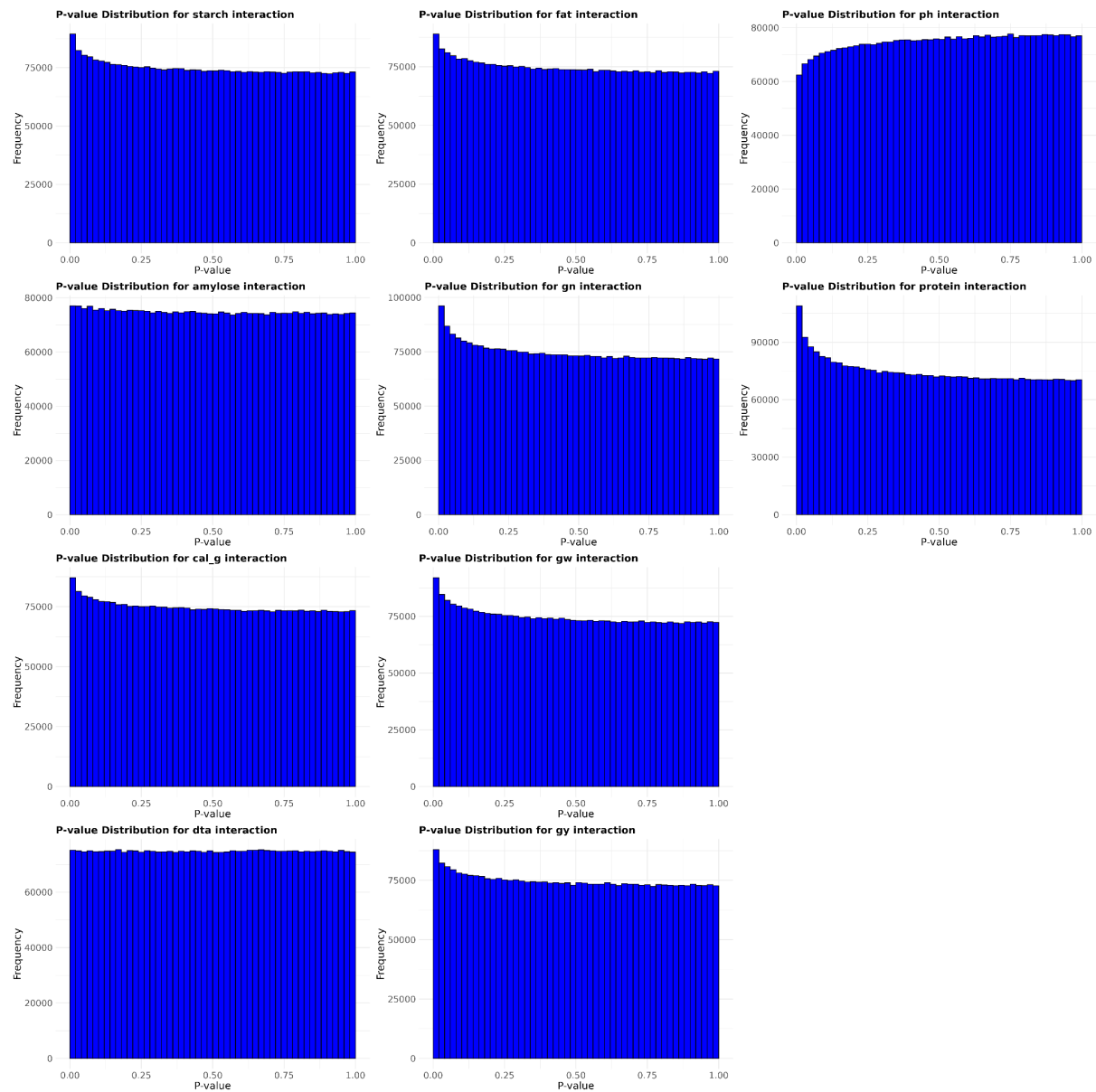
Lastly, understanding epistatic interactions can inform breeding programs by identifying combinations of variants that collectively influence key traits. While this project focused on computational predictions, the next step could potentially involve experimental validation of the most promising candidate interactions to establish their role in trait variability. Integrating these findings into breeding strategies has the potential to enhance crop yield, quality, and adaptability.

In conclusion, this project highlights the challenges of detecting epistasis in polygenic traits, particularly due to the large number of SNP combinations and small effect sizes. The findings emphasize the need for larger datasets and more efficient computational methods to improve the power of detection. Focusing on coding regions and optimizing the pipeline were important steps to make the analysis feasible, but future studies should expand our procedure to include non-coding regions and explore alternative models to better capture complex genetic interactions. As epistasis research evolves, further refinements and additional data will be crucial to fully unravel the role of genetic interactions in shaping complex traits.

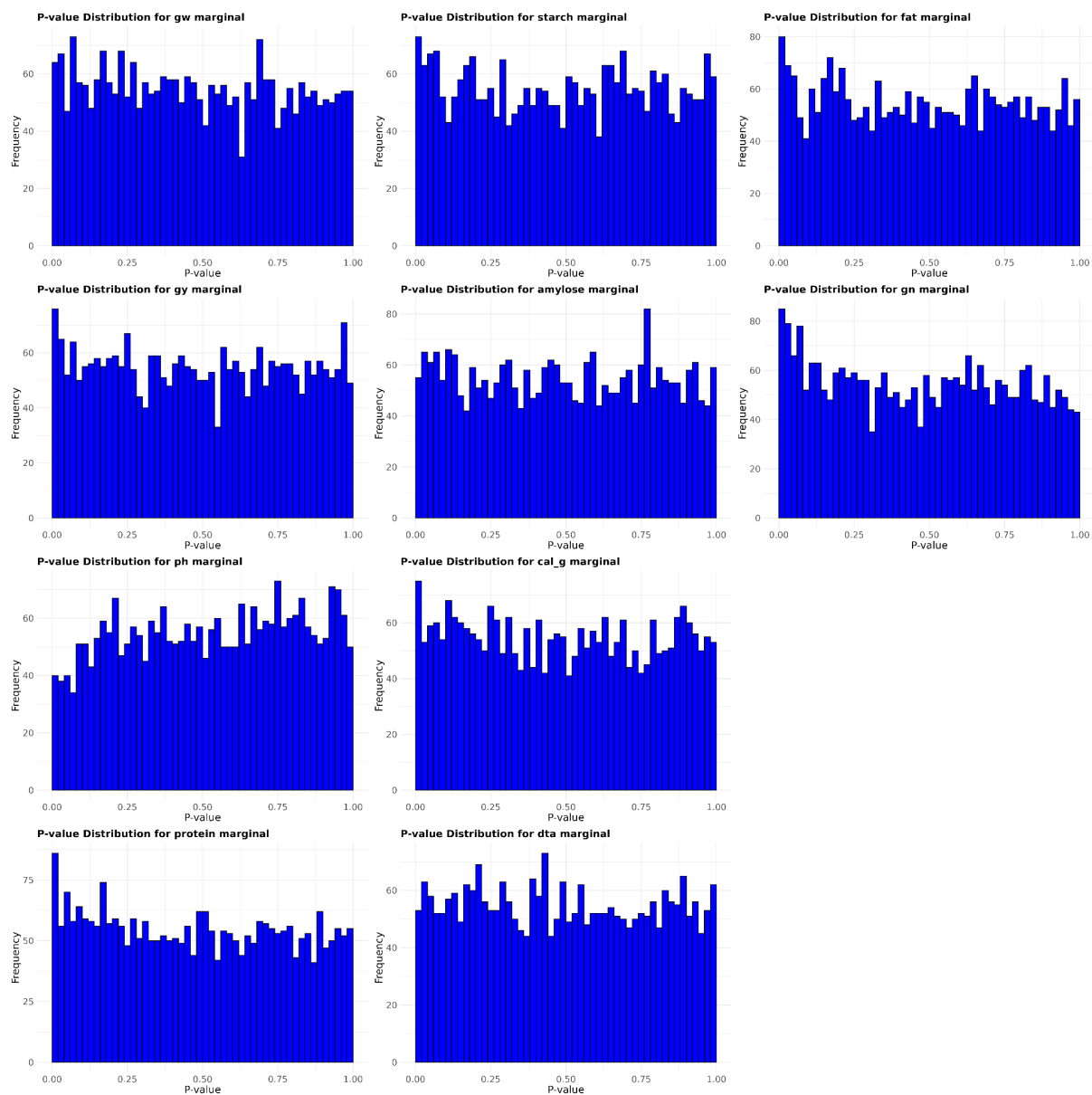
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Supplementary Material



Supplementary Figure 1. Distribution of raw interaction P-values across different traits



Supplementary Figure 2. Distribution of raw marginal P-values across different traits